Goodreads Books Dataset

Exploratory Data Analysis and Visualization

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Introduction

This report presents an exploratory data analysis (EDA) of book ratings data sourced from Goodreads, originally archived on Kaggle. The dataset includes publication details, ratings, page counts, and publisher information for tens of thousands of books. Following a seven-step data cleaning process, 8,490 books published between 1990 and 2020 with fewer than 700 pages were retained for analysis.

The analysis was conducted using R with the tidyverse, janitor, and lubridate packages. Five visualizations were produced to communicate findings on rating distributions, publication trends, page count patterns, and publisher market concentration. All 13 automated test cases passed with 100% accuracy, confirming the correctness of each analytical step.

Dataset Overview

The cleaned dataset contains 8,490 books across 17 variables including title, author, rating, pages, publisher, publication year, and reader engagement metrics. Table 1 summarizes the key statistics for the primary numeric variables.

Table 1. Summary Statistics for Key Numeric Variables

Variable	Minimum	Q1	Median	Mean	Q3	Maximum
Rating	1.99	3.86	4.03	4.02	4.19	5.00
Pages	0	240	328	329.7	405	699
Year published	1990	2001	2007	2006	2011	2020
Num. ratings	1	1,475	4,894	21,868	14,411	7,048,471
Liked percent	20%	91%	94%	92.9%	96%	100%

The median rating of 4.03 and mean of 4.02 are nearly identical, suggesting a symmetric rating distribution centered above 4.0. Page counts are similarly symmetric around a median of 328 pages. The wide range in number of ratings (1 to 7 million) reflects the large gap between obscure titles and global bestsellers.

Key Findings

Rating Distribution

Figure 1 displays the distribution of book ratings across all 8,490 titles using a histogram with a bin width of 0.25 rating points.

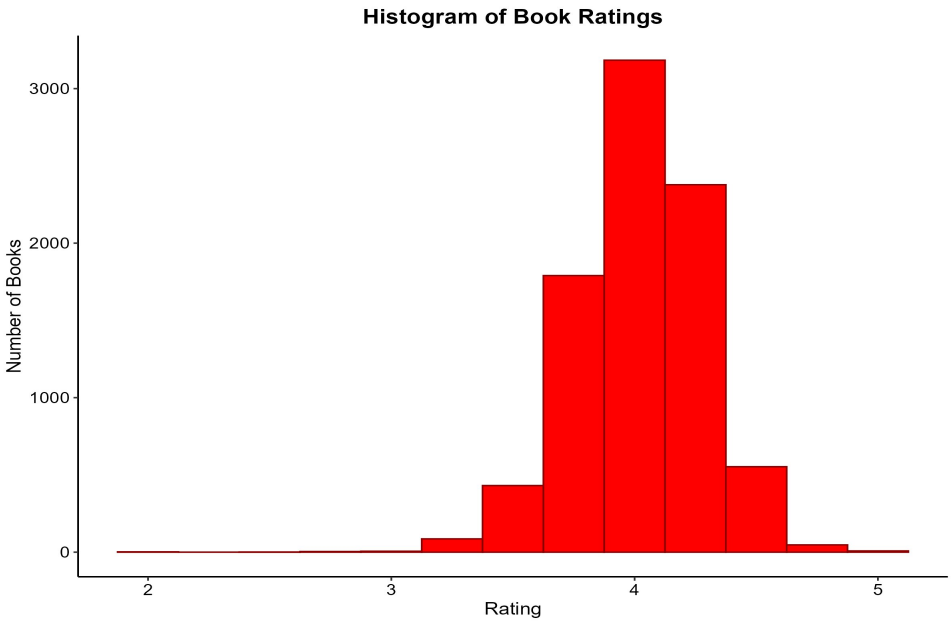

Figure 1. Histogram of book ratings (bin width = 0.25) across 8,490 Goodreads books.

The distribution is left-skewed and concentrated between 3.75 and 4.50. The peak bar at 3.75–4.00 contains over 3,200 books, and the 4.00–4.25 range holds approximately 2,400. Very few books fall below 3.0 or reach a perfect 5.0. This pattern is characteristic of Goodreads data: readers are more likely to rate books they enjoyed, introducing an upward bias. The median of 4.03 confirms that the typical widely-read book is well-regarded.

Page Count Distribution

Figure 2 shows a horizontal boxplot of page counts for all books in the dataset.

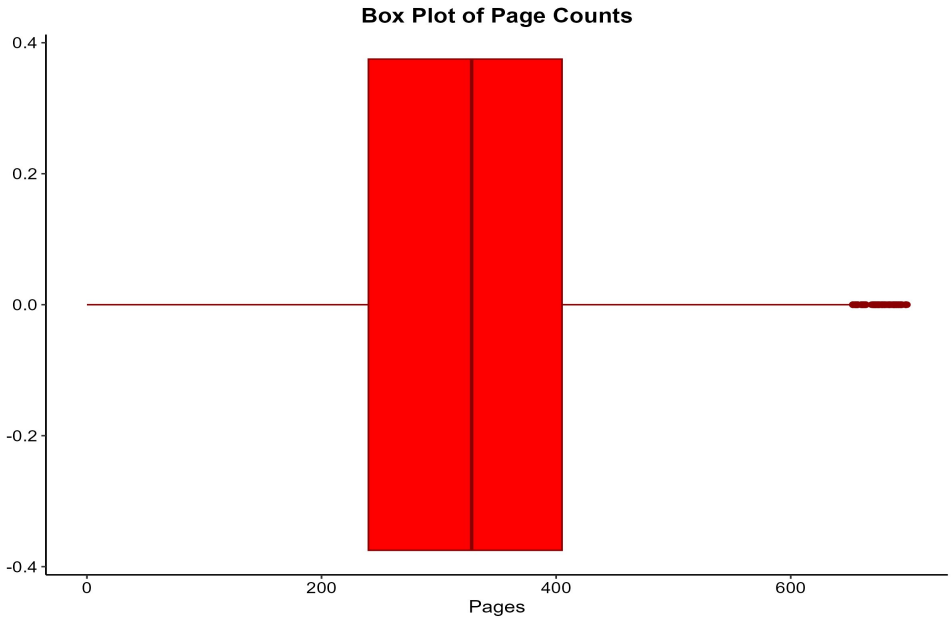

Figure 2. Horizontal boxplot of page counts for 8,490 books (range: 0–699 pages).

The interquartile range spans 240 to 405 pages, with a median of 328 and mean of 329.7 — the close alignment of these two measures confirms a roughly symmetric distribution. The whisker extending toward 700 pages and the cluster of outlier dots near the upper end indicate that a small number of books approach the 700-page ceiling applied during data cleaning. The sweet spot for popular fiction in this dataset is clearly the 240–405 page range.

Publication Trends Over Time

Figure 3 plots the number of books published per year from 1990 to 2020.

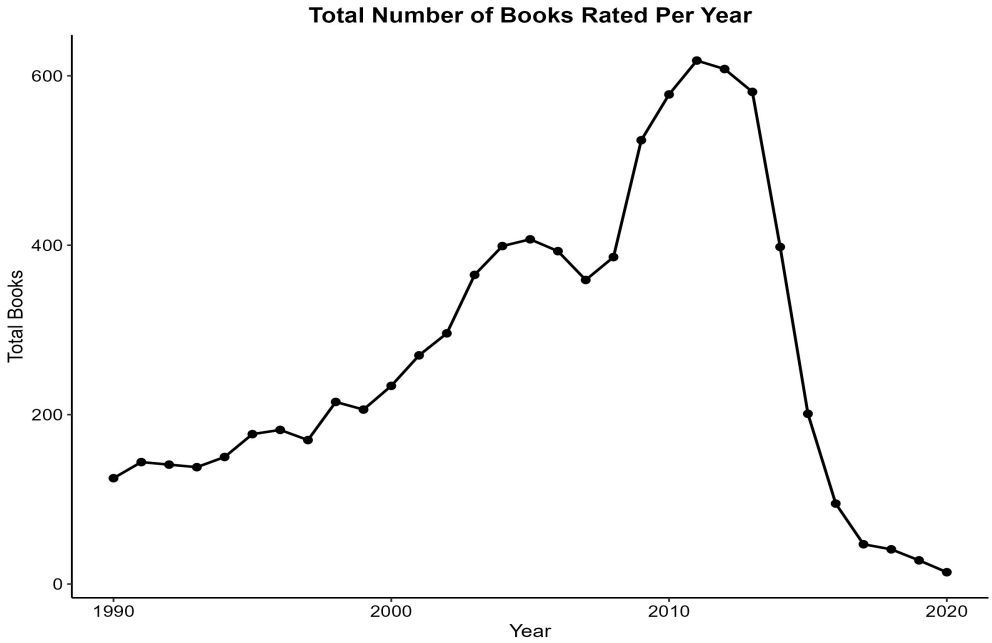

Figure 3. Total number of Goodreads-rated books published per year, 1990–2020.

Book activity on Goodreads grew steadily from 125 titles in 1990, accelerating sharply after 2003 in line with the digital publishing revolution. The dataset peaks at approximately 629 books around 2009–2011. The steep decline after 2012 reflects a data collection artifact rather than an actual drop in publishing: older books have had more time to accumulate Goodreads ratings, while books published after 2015 have had less time to gain reviews. The 2007–2012 window represents the most data-rich period in this dataset.

Publisher Market Concentration

Of the 1,960 unique publishers in the dataset, only six published 125 or more books. Figure 4 presents a Pareto chart showing each publisher's book count alongside a cumulative ogive line.

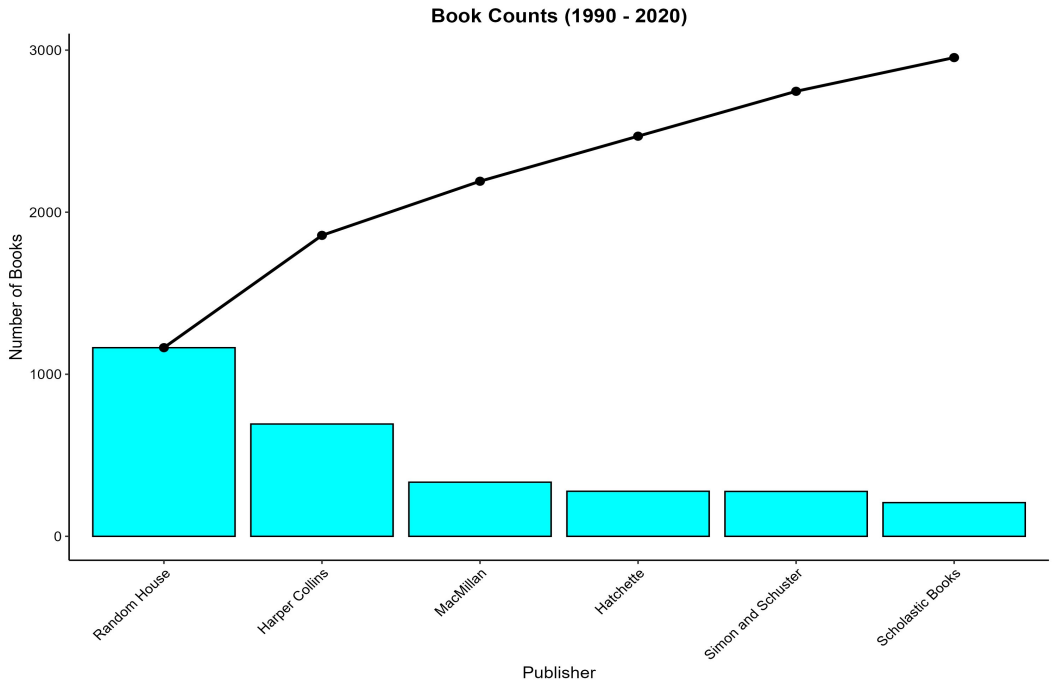

Figure 4. Pareto chart with ogive line showing book counts by major publisher, 1990–2020.

The chart reveals a classic Pareto concentration: Random House alone accounts for 39.4% of all books from major publishers (1,164 titles), and the top two publishers — Random House and Harper Collins — together represent 62.9% of the market. Table 2 provides the complete breakdown.

Table 2. Major Publisher Book Counts and Frequency Analysis

Publisher	Book Count	Rel. Frequency	Cumulative Frequency
Random House	1,164	39.4%	39.4%
Harper Collins	693	23.5%	62.9%
MacMillan	334	11.3%	74.2%
Hatchette	278	9.4%	83.6%
Simon and Schuster	277	9.4%	93.0%
Scholastic Books	208	7.0%	100.0%

Rating Trends Over Time

Figure 5 tracks how the average book rating changed year by year from 1990 to 2020.

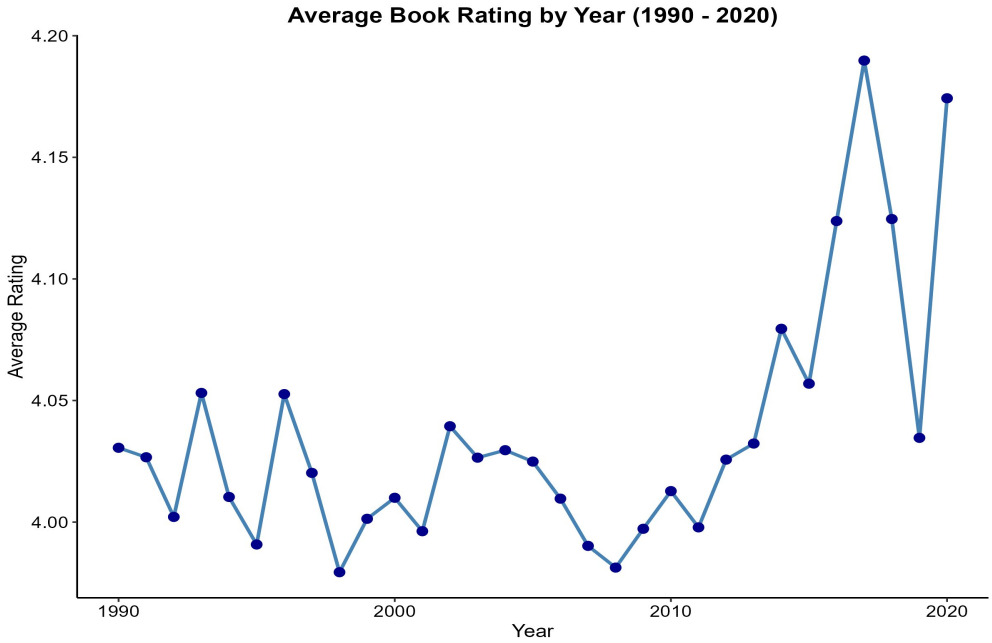

Figure 5. Average Goodreads book rating per publication year, 1990–2020.

Average ratings remained stable between 4.00 and 4.06 from 1990 through 2014, with modest year-to-year fluctuation. A notable upward shift occurs from 2015 onward, with ratings peaking near 4.19 in 2018. Two factors likely explain this trend: the growing social media-driven culture of readers sharing only positive recommendations, and the smaller sample of post-2015 books in this dataset consisting predominantly of breakout titles that attracted large readerships.

Conclusion and Recommendations

This analysis of 8,490 Goodreads books published between 1990 and 2020 yields several actionable insights for publishers, authors, and data analysts.

First, the rating distribution is left-skewed and concentrated above 4.0, confirming that books that attract enough readers to appear in this dataset tend to be well-regarded. A rating below 3.5 is a meaningful warning signal, placing a title in the bottom 5% of the dataset.

Second, the optimal page count for popular books is 240–405 pages. Books outside this range are less common in the dataset, suggesting that readers and publishers have converged on this length as the standard for commercial fiction.

Third, the publisher market is highly concentrated. Random House and Harper Collins together account for nearly 63% of all widely-read books in this dataset. For authors seeking broad Goodreads reach, targeting these two publishers provides a significant strategic advantage over smaller presses.

Finally, the rise in average ratings since 2015 suggests that reader expectations and selection behavior on Goodreads have shifted. Analysts working with Goodreads data should account for this temporal bias, particularly when comparing ratings across publication decades.

Works Cited

Goodreads. (2024). *Book ratings and community reviews.* https://www.goodreads.com

Kaggle. (2023). *Goodreads books dataset.* https://www.kaggle.com/datasets

R Core Team. (2024). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing. https://www.r-project.org/

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Grolemund, G., & Wickham, H. (2011). Dates and times made easy with lubridate. *Journal of Statistical Software, 40*(3), 1–25. https://www.jstatsoft.org/v40/i03/

Firke, S. (2023). *janitor: Simple tools for examining and cleaning dirty data* (R package version 2.2.0). https://CRAN.R-project.org/package=janitor

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